

Introduction

Given a material, its electrical conductivity gives a relation between the resistance of wire and its dimensions:

$$R = \frac{1}{\sigma} \times \frac{L}{A}$$

The thermal conductivity gives information about the rate at which heat is transferred to the material. For cross sectional area A , rate of heat transfer is given by:

$$\frac{dQ}{dt} = KA \frac{dT}{dx}$$

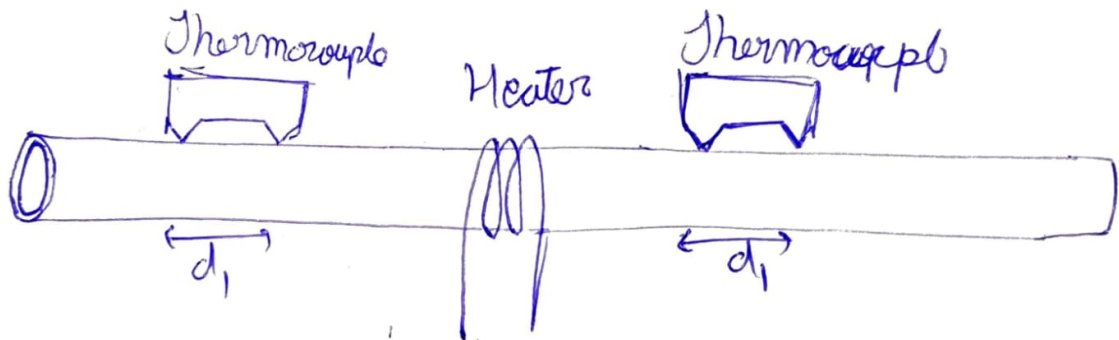
In this experiment, we determine the thermal and electrical conductivity of copper.

Experiment Setup and Procedure

Apparatus

Thermal and electrical conductivity box, DC current sources and a multichannel voltmeter connected to the box.

Experiment Setup



The thermal and electrical conductivity box contains a copper tube that is connected to a heater. On both sides of the heater, there is a thermocouple which gives reading to channels 1 and 2 respectively.

R = Heater Resistance = 22Ω

Outer diameter of copper tube $D = 9.5 \text{ mm}$

Wall Thickness of copper tube $t = 0.75 \text{ mm}$

Distance between thermocouple junctions $d_1 = 60 \text{ mm}$

Distance between Voltage Leads $d_2 = 180 \text{ mm}$

Thermo-electric power of Thermocouple: $\alpha = 40 \mu\text{V}/\text{K}$

Procedure

First, we measure the electrical conductivity. For different values of current, the voltage across the voltage leads is measured. Using this data and the dimensions of copper tube, we can determine electrical conductivity.

After this, we send different values of current through the heater. Since the resistance of heater is known, we know the heat released by the heater. We measure the potential difference between the the thermocouple leads. The system is allowed to reach equilibrium, then the potential difference across this thermocouple and the thermocouple on the other side are noted. Using this data, thermal conductivity is calculated.

Formulae and Relations

Outer radius of tube = $r_2 = D/2$

Inner radius of tube = $r_1 = r_2 - t$

$$A = \pi(r_2^2 - r_1^2)$$

$$R = \frac{V}{I}$$

$$\therefore I = \frac{\sigma A}{d_2} V$$

Upon plotting this, $slope = \frac{\sigma A}{d_2}$

$$\therefore \sigma = slope \times \frac{d_2}{A}$$

Voltage across first thermocouple = V_1

Voltage across second thermocouple = V_2

Temperature differences across thermocouples;

$$\Delta T_1 = V_1 / \alpha$$

$$\Delta T_2 = V_2 / \alpha$$

Power output by heater = $P = I^2 R$

Power output by heater = The heat being transferred by on both sides

$$\therefore I^2 R = KA(\Delta T_1 + \Delta T_2) / d_1$$

$$K = \frac{I^2 R d_1}{A(\Delta T_1 + \Delta T_2)}$$

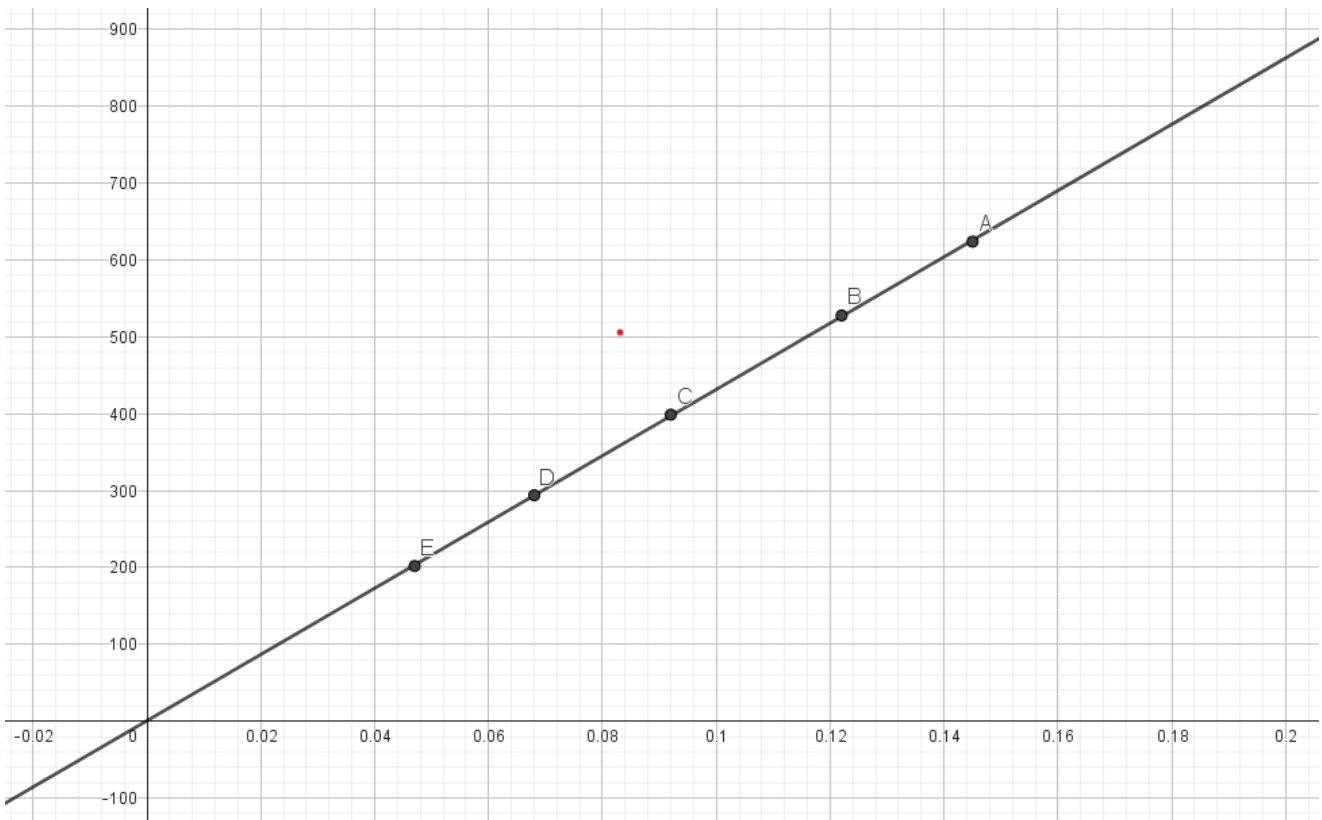
Data and Analysis

Using the formulae above, we get

$$A = 2.061 \times 10^{-5} m^2$$

For the first part of the experiment, the following data was collected:

$I \text{ (mA)}$	$V \text{ (mV)}$
624	0.145
528	0.122
399	0.092
294	0.068
202	0.047



$$y = 4311.16 x + 0.7$$

$$\therefore \sigma = slope \times \frac{d_2}{A} = 3.76 \times 10^7 S/m$$

For the second part of the experiment, the following values were noted:

$I \text{ (A)}$	$V_1(\mu V)$	$V_2(\mu V)$	ΔT_1	ΔT_2	$\Delta T_1 + \Delta T_2$	K
0.215	184	196	4.6	4.9	9.5	311.64
0.262	264	279	6.6	7.0	13.6	323.86
0.169	114	122	2.85	3.0	5.85	310.03

Taking the average value, we determine

$$K = 315.17 W/mK$$